Blueprint

The Pentagon, Machine-Checked: Abel's five-term relation for the dilogarithm in Lean 4

Special functions · the dilogarithm

Carles Marín

Statements

Theorem 1 — Dilog.rogersL_fiveterm 3
Lemma 1 — Dilog.hasDerivAt_rogersL 4
Lemma 2 — Dilog.tendsto_rogersL_nhdsGT_zero 4
Lemma 3 — Dilog.continuousAt_rogersL 4

At a glance

What the paper contributes

A machine-checked proof of the identity that generates the weight-2 theory of the dilogarithm, on the dilogarithm built in Lean by the companion paper.

- **Abel's five-term relation** (Theorem 1), sorry-free in Lean 4: for x, y in (0,1), L(x) + L(y) = L(xy) + L(x(1−y)/(1−xy)) + L(y(1−x)/(1−xy)), with L the Rogers L-function.
- **The classical route, made formal**: the five-term difference has zero derivative in x (nine logarithms collapse into a basis of five), so it is constant; its value is read off as x → 0⁺, where log t · log(1−t) → 0 by a squeeze.
- **Three reusable blocks**: the derivative of L on (0,1), L(t) → 0 as t → 0⁺, continuity of L on (0,1).
- The mathematics is classical (Spence, Abel, Rogers); the paper claims the formalization, after a documented search of proof-assistant libraries.

🚀 **Start here: Theorem 1**: Abel's five-term relation for the Rogers L-function on (0,1)².

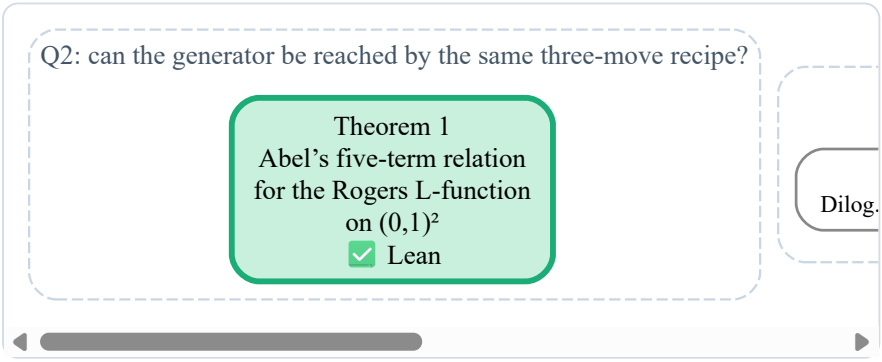
Main results:

- **Theorem 1**: Abel's five-term relation for the Rogers L-function on (0,1)²

4 results | ✅ 1 formalized in Lean | ◐ 0 with a part in Lean | 💻 0 checked by a program | written proof only | (a result can carry several tags)

🗺️ Map of the results (click a box to go to it)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



📐 Setting (copied from the paper, §1)

The cleanest statement uses the *Rogers* L*-function* $L(x) := \mathrm{Li}_2(x) + \frac{1}{2}\log x \log(1-x)$, which absorbs the stray logarithms of the raw Li_2 form and turns the relation into a pure sum of five L values. *(§2)* Fix $y \in (0,1)$ and set

$$G(t) = \mathrm{L}(t) - \mathrm{L}(ty) - \mathrm{L}\left(\tfrac{t(1-y)}{1-ty}\right) - \mathrm{L}\left(\tfrac{y(1-t)}{1-ty}\right).$$

The theorem is $G(x) = -\mathrm{L}(y)$.

2 Q2: can the generator be reached by the same three-move recipe?

Theorem 1. ✅ Lean (`Dilog.rogersL_fiveterm`)✓ 📕 p. 3

> *For all* $x, y \in (0,1)$,
>
> $$\mathrm{L}(x) + \mathrm{L}(y) = \mathrm{L}(xy) + \mathrm{L}\left(\tfrac{x(1-y)}{1-xy}\right) + \mathrm{L}\left(\tfrac{y(1-x)}{1-xy}\right).$$

✅ Checked when this page was built: FiveTerm.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — rogersL_fiveterm, fiveterm_hasDerivAt_zero, hasDerivAt_rogersL, tendsto_rogersL_nhdsGT_zero, continuousAt_rogersL: propext, Classical.choice, Quot.sound. Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean FiveTerm.lean` (SHA-256 of the compiled file: `ce29fa490a366ddb…`; a source page on this site may show it with comments edited, under its own hash).

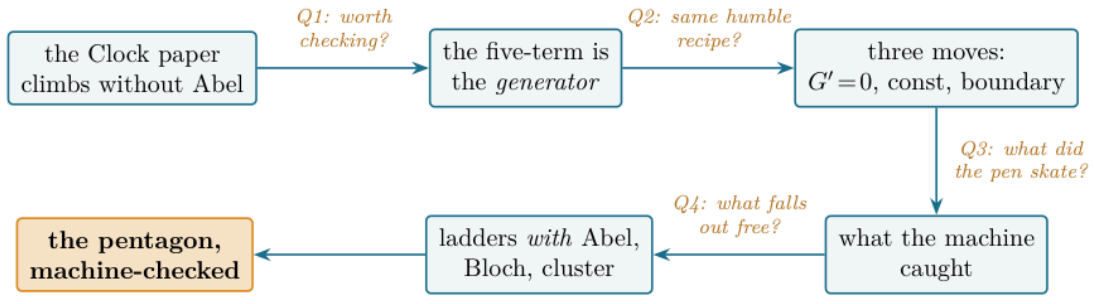
> 🆕 **Classical and new:** classical (Spence, Abel, Rogers); the paper claims the formalization: «to the best of my knowledge the first machine-checked five-term relation, on the first machine-checked dilogarithm, in any proof assistant» (§6), from a documented search of June 2026.

↳ Not used by another result of the paper.

Proof

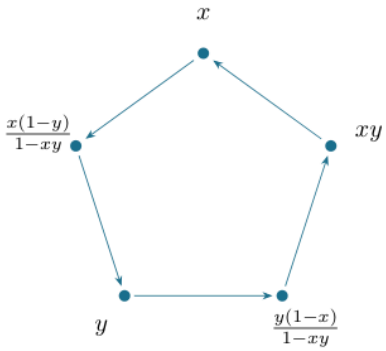
> Formalized in Lean (see the linked declarations).

🖼️ **Figure 1**



The reader's map. The thread starts where the companion *Clock* paper left it (the golden ladder climbed without Abel) and closes at the machine-checked pentagon; each arrow is the question its section answers. Every node is a block of `sorry`-free Lean.

Figure 2



The five arguments on a cycle. The five-fold symmetry of this pentagon is the combinatorial shadow shared by the Bloch-group relation, the cluster-algebra mutation pentagon, and the quantum-dilogarithm pentagon identity.

5 The building blocks

Lemma 1. (`Dilog.hasDerivAt_rogersL`) p. 6

> *On* $(0,1)$, $\mathrm{L}'(x) = -\frac{1}{2}\left(\frac{\log(1-x)}{x} + \frac{\log x}{1-x}\right)$.

↳ Not used by another result of the paper.

Lemma 2. (`Dilog.tendsto_rogersL_nhdsGT_zero`) p. 6

> $\mathrm{L}(t) \to 0$ *as* $t \to 0^{+}$.

↳ Not used by another result of the paper.

Lemma 3. (`Dilog.continuousAt_rogersL`) p. 6

> L *is continuous at every* $x \in (0,1)$.

↳ Not used by another result of the paper.

References

1

N. H. Abel, *Œuvres complètes* (L. Sylow, S. Lie, eds.), vol. II, Grøndahl, Christiania, 1881; the five-term relation is in the note on the dilogarithm. 🔎 find it

2

L. J. Rogers, On function sum theorems connected with the series $\sum x^n/n^2$, *Proc. London Math. Soc.* **4** (1907) 169–189. 🔎 find it

3

L. Lewin, *Polylogarithms and Associated Functions*, North-Holland, 1981. 🔎 find it

4

D. Zagier, The dilogarithm function, in *Frontiers in Number Theory, Physics, and Geometry II*, Springer, 2007, 3–65. 🔎 find it

5

Z. Wojtkowiak, The basic structure of polylogarithmic functional equations, in *Structural Properties of Polylogarithms* (L. Lewin, ed.), Math. Surveys Monogr. **37**, AMS, 1991, 205–231. find it

6

R. de Jeu, Towards regulator formulae for the K-theory of curves and number fields, *Compositio Math.* **124** (2000) 137–194. find it

7

S. Bloch, *Higher Regulators, Algebraic K-theory, and Zeta Functions of Elliptic Curves*, CRM Monogr. Ser. **11**, AMS, 2000 (Irvine lectures, 1978). find it

8

A. A. Suslin, K_3 of a field, and the Bloch group, *Proc. Steklov Inst. Math.* **183** (1991) 217–239. find it

9

W. D. Neumann, D. Zagier, Volumes of hyperbolic three-manifolds, *Topology* **24** (1985) 307–332. find it

10

A. B. Goncharov, Geometry of configurations, polylogarithms, and motivic cohomology, *Adv. Math.* **114** (1995) 197–318. find it

11

T. Nakanishi, Dilogarithm identities for conformal field theories and cluster algebras, in *Symmetries, Integrable Systems and Representations*, Springer Proc. Math. Stat. **40** (2013) 407–444. find it

12

L. D. Faddeev, R. M. Kashaev, Quantum dilogarithm as a kernel of the quantum modular double, *Modern Phys. Lett. A* **9** (1994) 427–434. find it

13

A. N. Kirillov, Dilogarithm identities, *Prog. Theor. Phys. Suppl.* **118** (1995) 61–142. find it

14

Al. B. Zamolodchikov, Thermodynamic Bethe ansatz in relativistic models, *Nucl. Phys. B* **342** (1990) 695–720. find it

15

W. Nahm, Conformal field theory and torsion elements of the Bloch group, in *Frontiers in Number Theory, Physics, and Geometry II*, Springer, 2007, 67–132. find it

16

C. Marín, *The Clock That Never Ticks: a machine-checked path from the dilogarithm to quantum speed limits in Lean 4*, Zenodo, 2026. doi:10.5281/zenodo.20675271.

17

The mathlib Community, The Lean mathematical library, in *CPP 2020*, ACM, 2020, 367–381. 🔎 find it